



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

Programme Name & Branch : BTECH – Computer Science and Engineering
Course Code and Course Name : BCSE313L – Fundamentals of Fog and Edge Computing
Faculty Name(s) : Prof. Ramanathan L/Prof. Naveenkumar L/
Prof. Badrinath N /Prof. Kauser Ahmed P/
Prof. Justin Gopinath A/Prof. Munusamy S.
Class Number(s) : VL2024250501566/01570/01577/01599/01602/06625
Date of Examination : 01/02/2025
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes: CO1- Explore technologies behind the communication and management of fogs and edge resources.CO4-Analyze the performance and issues of the applications developed using fog and edge architecture.CO2-Learn the techniques for storage and computation in fogs, edges, 5G and clouds.

Q. No	Question	M	CO	BL
1.	Describe the business models of Fog and edge computing platforms with suitable examples. Answer key: The three basic business models are 1. X as a Service 2. Support Service 3. Application Service 1.X as a Service XaaS providers of FEC allow their clients to pay to use the hardware equipment that supports SCANC mechanisms 1. SaaS (Software as a Service): This is probably the most common type of XaaS. It involves using software applications delivered over the internet on a subscription basis. Examples include: • Microsoft Office 365: Instead of buying a physical copy of Microsoft Word or Excel, you can subscribe to Office 365 and access the software programs online. • Dropbox: This cloud storage service allows you to store your files online and access them from any device. • Netflix: This streaming service provides access to a vast library of movies and TV shows for a monthly subscription fee. 2. PaaS (Platform as a Service): This type of XaaS provides developers with a platform to build and deploy their applications without managing the underlying infrastructure. Here are some examples:	10	1	1



**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025**

SLOT: F1+TF1

	• Amazon Web Services (AWS) Lambda: This serverless computing platform lets developers write and run code without provisioning or managing servers. • Google App Engine: This platform allows developers to build and deploy web applications on Google's infrastructure. 3. IaaS (Infrastructure as a Service): This XaaS model offers access to virtual machines, storage, and networking resources that businesses can use to build their own IT infrastructure in the cloud. Some examples include: • Microsoft Azure: This cloud computing platform provides a range of infrastructure services, including virtual machines, storage, and networking. • Google Cloud Platform: Similar to Azure, it offers a suite of cloud computing services for businesses to build and deploy applications. 2.Support Service The support service of FEC is similar to the software management support services in general information systems in which the clients who own the hardware equipment can pay the support service provider to provide them the corresponding software installation, configuration, and updates on the clients' equipment based on the requirements of the clients. • Example: – Cisco provides the fog computing solution, in which the clients purchase Cisco's IOX-enabled equipment and then pay an additional service fee to gain access to the software update and technical support for configuring their FEC environments 3.Application Service Application service providers provide application solutions to help their clients in processing the data within or outside of the client's operation environments Example: • An IndieFog provider who has installed Apache Edgent can offer the built-in stream data classification function as an application service for the mobile AAL(ambient assisted living) clients in the close proximity.			
2.	Design an edge-based Smart health monitoring system that collects vital signs data (heart rate, oxygen saturation). The system should detect potential health risks and send alerts to users and emergency services. (a) Illustrate and explain the suitable diagram for fog edge computing paradigm? (4 M) (b) Compare how the above paradigm CIOT approach? (6 M) Answer Key: The system continuously collects vital signs data (heart rate, oxygen saturation) from wearable sensors. It processes this data locally at the edge to detect potential health risks in real-time and sends alerts to users and emergency services when necessary. Architecture diagram 1. Wearable Sensors: These are the primary data collection points, capturing heart rate and oxygen saturation data.	10	1	4



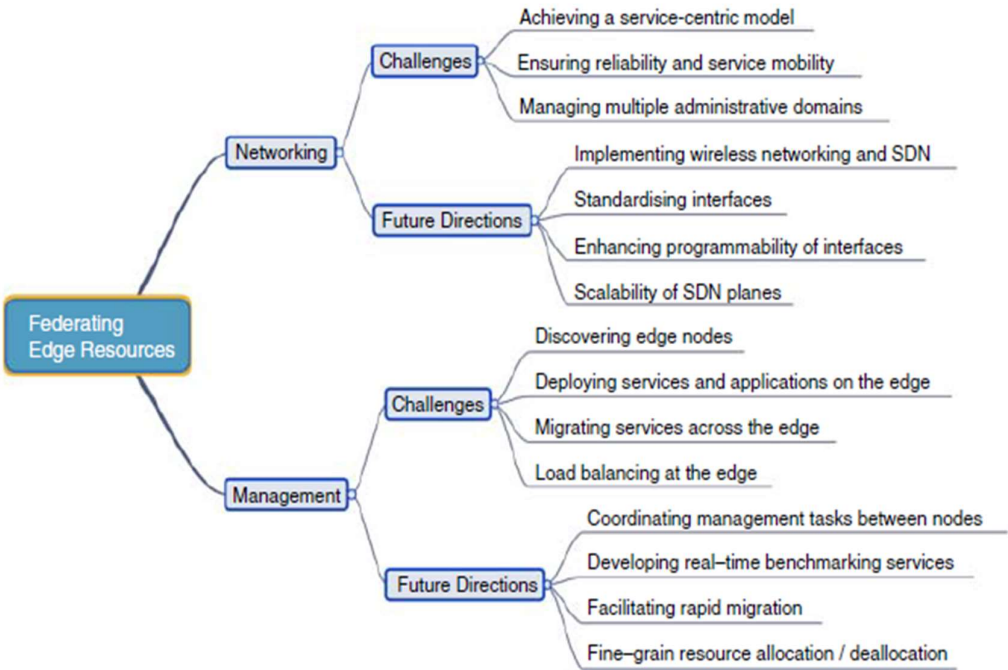
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

	2. Edge Computing Layer: This layer handles real-time data collection and processing, detecting immediate health risks and generating alerts. 3. Fog Computing Layer: Processes aggregated data from multiple edge devices, performs initial health risk detection, and filters data before sending it to the cloud. 4. Cloud Layer: Provides long-term storage, advanced analytics, and historical data analysis. Comparison with CIIoT Approach 1. Data Processing Fog Edge Computing: Processes data locally at the edge and fog layers, reducing latency and bandwidth usage. CIIoT (Cloud IIoT): Primarily relies on cloud-based processing, which can introduce higher latency due to data transmission to and from the cloud. 2. Latency Fog Edge Computing: Low latency due to local processing at the edge. CIIoT: Higher latency as data must travel to the cloud for processing. 3. Reliability Fog Edge Computing: More reliable for real-time applications since it can operate even with intermittent internet connectivity. CIIoT: Dependent on continuous and stable internet connectivity. 4. Bandwidth Usage Fog Edge Computing: Optimizes bandwidth usage by processing data locally and only sending relevant information to the cloud. CIIoT: Higher bandwidth usage due to continuous data transmission to the cloud. 5. Scalability Fog Edge Computing: Can be more scalable as processing is distributed across edge and fog nodes. CIIoT: Scalability might be limited by the cloud's capacity and network bandwidth. 6. Security Fog Edge Computing: Provides enhanced security by processing sensitive data locally and reducing exposure to the cloud. <td></td><td></td><td></td>			
--	--	--	--	--



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

	○ CIoT: Data is more exposed due to frequent transmission to and from the cloud, potentially increasing security risks.			
3.	(a) Elaborate the different network and management challenges in federated environment? 	5	4	6
	(b) A smart home security system uses edge computing to analyse video feeds from security cameras. The system consists of motion detection software, Video feeds from cameras, a local fog device running the program and storing the data. Using proper equation explain the overall reliability of the edge fog computing environment in the given scenario depends on the reliability of each component. Answer key: To explain the overall reliability of the edge fog computing environment in the given scenario, we can use the concept of series reliability. In a series system, the overall reliability of the system depends on the reliability of each individual component. If any component fails, the entire system can fail. R. K. Behera et al. (Reliability modelling of service-oriented Internet of Things) proposed an analytical modeling for estimating the reliability of an IoT scenario for a smart home context. • An algorithm is proposed to estimate the reliability of the IoT service, which is formed by n subsystems. • Reliability of IoT system is calculated using the below equation $R_s(t_b) = \prod_{i=1}^N ISR_i \times \prod_{i=1}^f P_f(i) \times \prod_{i=1}^k P_{pr}(i)$	5		



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

	(P_{pr}) the availability of the all k programs running on the virtual machines (P_f) the availability of the f input files for the programs (ISR) the reliability of each subsystem , i.e. the reliability of the VM being executed.			
4.	A city's surveillance system uses IP cameras to capture and analyse footage in real time. The system utilises edge computing to perform initial video processing and inject detection at the edge of the network reducing latency. The processed data is then transmitted to a local cloudier for further analysis such as facial recognition and license plate detection. Finally, the analysed data is sent to cloud for long term storage. Illustrate with a neat diagram how network slicing can be used to prioritise and manage different types of traffic in the smart surveillance system such as video feeds, alarm signals and software updates. Answer key: Diagram Physical Infrastructure: This represents the underlying shared network infrastructure (e.g., routers, switches, fiber optic cables). It's the foundation upon which the virtual networks are built. Network Slicing: This is the core concept. The physical infrastructure is divided into multiple isolated virtual networks called "slices." Each slice is allocated dedicated resources (bandwidth, latency, processing power) according to its specific needs. An orchestration layer manages this slicing process. Virtual Networks (Slices): Three slices are shown in this example: Slice 1 (Video Feed): This slice is optimized for high bandwidth and low latency, crucial for real-time video streaming from the IP cameras to the edge computing devices. It prioritizes the continuous flow of video data. Slice 2 (Alarm Signals): This slice is optimized for low bandwidth but <i>extremely</i> low latency. Alarm signals (e.g., triggered motion detectors) need to be transmitted instantly, even if the amount of data is small. Reliability is paramount here. Slice 3 (Software Updates): This slice is for software updates to the cameras or edge devices. It requires variable bandwidth (depending on the update size) and is generally "best effort." While important, a slight delay in software updates is usually acceptable compared to video feeds or alarms.	10	2	3



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT: F1+TF1

	4. Edge Computing (Pre-processing): IP cameras send video feeds to edge computing devices. These devices perform initial processing (e.g., basic motion detection, compression) to reduce the amount of data sent to the local cloud. 5. Local Cloud (Facial/License Recognition): The processed data is then sent to the local cloud for more intensive analysis like facial recognition and license plate detection. 6. Cloud Storage (Long Term): Finally, the analyzed data is sent to the cloud for long-term storage and potential later analysis. 7. Cloud Platform (Management): A cloud platform might be used to manage the entire system, including the network slices, resource allocation, and application deployment. How Network Slicing Helps: • Prioritization: Slicing ensures that critical traffic (like alarm signals) gets the highest priority and is guaranteed the necessary resources, even during periods of high network load. • Resource Allocation: Each slice gets the appropriate amount of bandwidth and other resources. The video feed slice gets high bandwidth, while the alarm signal slice gets low latency. • Isolation: The slices are isolated from each other. A surge in video traffic won't affect the transmission of alarm signals. • Flexibility: Slicing allows the network to adapt to changing needs. If more cameras are added, the video feed slice can be scaled up. • QoS (Quality of Service): Network slicing enables different QoS levels for different types of traffic, ensuring optimal performance for each.			
5.	a) What is the important role of SDN in Cloud computing? Answer key: Software-Defined Networking (SDN) plays a crucial role in cloud computing by providing a more flexible, agile, and efficient way to manage and control network resources. Here are some of the key ways SDN contributes to cloud computing: Dynamic Resource Allocation: SDN allows for on-demand allocation and provisioning of network resources (bandwidth, virtual networks, security policies) to meet the dynamic needs of cloud applications. This eliminates the need for manual configuration of network devices, enabling rapid scaling and adaptation to changing workloads. Programmable Network Control: SDN's centralized control plane allows administrators to programmatically manage network behavior through software APIs. This enables automation	5	2	1



**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025**

of network tasks, such as provisioning, configuration, and troubleshooting, leading to faster service deployment and reduced operational overhead. Network Virtualization: SDN enables the creation of virtual networks (VLANs) on top of the physical infrastructure, allowing for flexible partitioning and isolation of network resources for different cloud tenants or applications. This supports multi-tenancy and enables efficient resource utilization. Scalable Control Plane: SDN controllers can manage a large number of network devices and virtual networks, providing the scalability needed to support massive cloud deployments. As cloud environments grow, SDN can easily adapt and scale the network infrastructure to meet the increasing demands. Centralized Security Policy Enforcement: SDN enables centralized management and enforcement of security policies across the entire network. This simplifies security management and ensures consistent security posture across the cloud environment.			
b) Elaborate with example how SDN and NFV compliment network slicing? Answer key: SDN and NFV are complementary technologies that work together to enable and enhance network slicing. Here's how they complement each other with an example: Scenario: Imagine a 5G network operator wants to offer different network services with varying requirements: • Enhanced Mobile Broadband (eMBB): High bandwidth for video streaming and downloads. • Ultra-Reliable Low-Latency Communications (urLLC): Extremely low latency for autonomous vehicles and industrial automation. • Massive Machine-Type Communications (mMTC): Large number of connected devices with low bandwidth for IoT applications. How SDN and NFV Enable Network Slicing: 1. NFV (Network Function Virtualization): ○ Traditionally, network functions (like firewalls, load balancers, and routing) were implemented on dedicated hardware appliances. ○ NFV virtualizes these network functions, allowing them to run as software on general-purpose servers. ○ In our example, the 5G core network functions (like mobility management, session management, and policy control) are virtualized using NFV. 2. SDN (Software-Defined Networking):	5		



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

	○ SDN separates the control plane (which makes decisions about network traffic) from the data plane (which forwards the traffic). ○ SDN provides a centralized controller that can programmatically control the network infrastructure (switches, routers, etc.) through software APIs. ○ In our example, the SDN controller manages the virtualized network functions and the underlying physical network infrastructure. 3. Network Slicing: ○ Using NFV, the operator can create multiple virtual instances of network functions, each tailored to the specific requirements of a service. ○ Using SDN, the operator can then allocate resources (bandwidth, computing power, storage) to these virtual network functions and create isolated network slices. ○ In our example, the operator creates three network slices: ▪ eMBB slice: Virtualized network functions optimized for high bandwidth and throughput. ▪ urLLC slice: Virtualized network functions optimized for ultra-low latency and high reliability. ▪ mMTC slice: Virtualized network functions optimized for a large number of connections and low power consumption. How SDN and NFV Complement Network Slicing: • NFV enables the creation of virtualized network functions, which are the building blocks of network slices. • SDN provides the control and orchestration capabilities to manage these virtualized network functions and allocate resources to different slices. • Together, SDN and NFV enable the dynamic creation and management of network slices, allowing the operator to offer customized network services with specific performance characteristics. Example: • A user wants to stream a 4K video on their mobile device. The network operator assigns their connection to the eMBB slice, which provides the necessary bandwidth and throughput for a smooth video streaming experience.			
--	--	--	--	--



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

	• An autonomous vehicle needs to communicate with a traffic management system. The connection is assigned to the urLLC slice, which guarantees ultra-low latency and high reliability for critical control signals. • A smart city deploys thousands of sensors to monitor environmental conditions. The sensors are connected to the mMTC slice, which supports a massive number of devices with low bandwidth requirements.			
--	---	--	--	--
